

Objectives

The primary objective of this stage is to prepare the validated dataset for Supervised Learning by loading the `clean_air_quality_data.csv` and utilizing a representative sample of 500,000 rows. This sample size ensures computational efficiency without sacrificing statistical significance. The implementation defines `pm2_5` as the continuous numerical Target (Y), while `co`, `no2`, and `so2` serve as the Features (X), providing the chemical and temporal fingerprint necessary for environmental estimation.

Tasks Completed

1. Preparation for Hyperparameter tuning — In order to minimize long waiting time, 10% of the training data was created and used as a sample for the initial hyperparameter search.
2. Implementation of Randomized Search — `RandomizedSearchCV` was employed to explore a defined parameter space. The following distributions were used:

```
param_dist = {  
    'n_estimators': [50, 100],  
    'max_depth': [10, 20, None],  
    'min_samples_split': [5, 10],  
    'bootstrap': [True]  
}
```

3. Best Parameter Identification — the best hyperparameters were found using the sample defined:

```
Success! Best Parameters Found: {'n_estimators': 50, 'min_samples_split': 10, 'max_depth': None, 'bootstrap': True}  
-----
```

4. Full Model Training — A new `RandomForestRegressor` was initialized with the best parameters found during the randomized search.
5. Model Training on full dataset — The tuned model was then trained on the complete `X_train` and `y_train` datasets so that it learned from all available training data.
6. Performance Evaluation — Predictions were generated on the `X_test` dataset (from the preprocessing phase), then the performance of the tuned model was evaluated using Mean Absolute Error (MAE) and R-squared metrics.

Tasks to be Completed

- Possibility for further refinement on the hyperparameter search space or method.

Results and Analysis

The optimal hyperparameters were identified based on the sample data:

```
Success! Best Parameters Found: {'n_estimators': 50, 'min_samples_split': 10, 'max_depth': None, 'bootstrap': True}
```

Using these parameters, the tuned model achieved the following performance on the test set:

```
-----  
Tuned Mean Absolute Error: 5.3445  
Tuned R-squared Score: 0.5237  
-----
```

This does not demonstrate a significant increase in accuracy but it signifies that the hyperparameter selection still has room for growth. A refined version is recommended for the next development of the model with the use of targeted and more focused range for hyperparameters search once promising regions are identified.

Challenges and Solutions

Challenge: The primary challenge in this phase was the long training time for RandomizedSearchCV which my current available device is incapable of reducing significantly.

Solution: Used a 10% sample of the training data to reduce the overall computation complexity and time.

Conclusion

The Fine-Tuning phase successfully achieved its objective of improving the Random Forest Regressor's performance, albeit on a slight scale. By leveraging RandomizedSearchCV and efficiently managing computational resources through data sampling, we identified a set of hyperparameters that led to a slightly more accurate model, as evidenced by the reduced MAE and increased R-squared score. The tuned model now provides a stronger foundation for subsequent evaluation, demonstrating the value of systematic optimization.